Working with Data in Python Cheat Sheet

Reading and writing files

Package/Method	Description	Syntax and Code Example
File opening modes	Different modes to open files for specific operations.	Syntax: r (reading) w (writing) a (appending) + (updating: read/write) b (binary, otherwise text) `Examples: with open("data.txt", "r") as file: content = file.read() print(content) with open("output.txt", "w") as f`

File reading methods	Different methods to read file content in various ways.	Syntax:
		`file.readlines() # reads all lines as a list`
		`readline() # reads the next line as a string`
		`file.read() # reads the entire file content as a string`
		Example:
		`with open("data.txt", "r") as file:`
		`  lines = file.readlines()`
		`  next_line = file.readline()`
		`  content = file.read()`

File writing methods	Different write methods to write content to a file.	Syntax:
		`file.write(content) # writes a string to the file`
		`file.writelines(lines) # writes a list of strings to the file`
		Example:
		`lines = ["Hello\n", "World\n"]`
		`with open("output.txt", "w") as file:`
		`  file.writelines(lines)`

Iterating over lines	Iterates through each line in the file	Syntax: `for line in file: # Code to process each line`

	using a `loop`.	Example:
		`with open("data.txt", "r") as file:`
		`for line in file: print(line)`

Open() and close()	Opens a file, performs operations, and explicitly closes the file using the close() method.	Syntax:
		`file = open(filename, mode) # Code that uses the file`
		`file.close()`
		Example:
		`file = open("data.txt", "r")`
		`content = file.read()`
		`file.close()`

with open()	Opens a file using a with block, ensuring automatic file closure after usage.	Syntax:
		`with open(filename, mode) as file: # Code that uses the file`
		Example:
		`with open("data.txt", "r") as file:`
		`content = file.read()`

Pandas

Package/Method	Description	Syntax and Code Example
.read_csv()	Reads data from a `.CSV` file and creates a DataFrame.	Syntax: dataframe_name = pd.read_csv("filename.csv") Example: df = pd.read_csv("data.csv")

.read_excel()	Reads data from an Excel file and creates a DataFrame.	Syntax:
		`dataframe_name = pd.read_excel("filename.xlsx")`
		Example:
		`df = pd.read_excel("data.xlsx")`

.to_csv()	Writes DataFrame to a CSV file.	Syntax:
		`dataframe_name.to_csv("output.csv", index=False)`
		Example:
		`df.to_csv("output.csv", index=False)`

Access Columns	Accesses a specific column using [] in the DataFrame.	Syntax:
		`dataframe_name["column_name"] # Accesses single column`
		`dataframe_name[["column1", "column2"]] # Accesses multiple columns`
		Example:
		`df["age"]`
		`df[["name", "age"]]`

describe()	Generates statistics summary of numeric columns in the DataFrame.	Syntax: `dataframe_name.describe()`

		Example: `df.describe()`

drop()	Removes specified rows or columns from the DataFrame. axis=1 indicates columns. axis=0 indicates rows.	Syntax:
		`dataframe_name.drop(["column1", "column2"], axis=1, inplace=True)`
		`dataframe_name.drop(index=[row1, row2], axis=0, inplace=True)`
		Example:
		`df.drop(["age", "salary"], axis=1, inplace=True) # Will drop columns`
		`df.drop(index=[5, 10], axis=0, inplace=True) # Will drop rows`

dropna()	Removes rows with missing NaN values from the DataFrame. axis=0 indicates rows.	Syntax:
		`dataframe_name.dropna(axis=0, inplace=True)`
		Example:
		`df.dropna(axis=0, inplace=True)`

duplicated()	Duplicate or repetitive values or records within a data set.	Syntax: `dataframe_name.duplicated()`

		Example: `duplicate_rows = df[df.duplicated()]`

Filter Rows	Creates a new DataFrame with rows that meet specified conditions.	Syntax:
		`filtered_df = dataframe_name[(Conditional_statements)]`
		Example:
		`filtered_df = df[(df["age"] > 30) & (df["salary"] < 50000)`

groupby()	Splits a DataFrame into groups based on specified criteria, enabling subsequent aggregation, transformation, or analysis within each group.	Syntax:
		`grouped = dataframe_name.groupby(by, axis=0, level=None, as_index=True, sort=True, group_keys=True, squeeze=False, observed=False, dropna=True)`
		Example:
		`grouped = df.groupby(["category", "region"]).agg({"sales": "sum"})`

head()	Displays the first n rows of the DataFrame.	Syntax: `dataframe_name.head(n)`

		Example: `df.head(5)`
Import pandas	Imports the Pandas library with the alias pd.	Syntax: `import pandas as pd` Example: `import pandas as pd`

info()	Provides information about the DataFrame, including data types and memory usage.	Syntax:
		`dataframe_name.info()`
		Example:
		`df.info()`

merge()	Merges two DataFrames based on multiple common columns.	Syntax:
		`merged_df = pd.merge(df1, df2, on=["column1", "column2"])`
		Example:
		`merged_df = pd.merge(sales, products, on=["product_id", "category_id"])`

print DataFrame	Displays the content of the DataFrame.	Syntax:
		`print(df) # or just type df`
		
		Example:
		`print(df)`
		`df`

replace()	Replaces specific values in a column with new values.	Syntax:
		`dataframe_name["column_name"].replace(old_value, new_value, inplace=True)`
		
		Example:
		`df["status"].replace("In Progress", "Active", inplace=True)`

tail()	Displays the last n rows of the DataFrame.	Syntax:
		`dataframe_name.tail(n)`
		
		Example:
		`df.tail(5)`

Numpy

Package/Method	Description	Syntax and Code Example
Importing NumPy	Imports the NumPy library.	Syntax: `import numpy as np` Example: `import numpy as np`

np.array()	Creates a one or multi-dimensional array,	Syntax:
		`array_1d = np.array([list1 values]) # 1D Array`
		`array_2d = np.array([[list1 values], [list2 values]]) # 2D Array`
		Example:
		`array_1d = np.array([1, 2, 3]) # 1D Array`
		`array_2d = np.array([[1, 2], [3, 4]]) # 2D Array`

Numpy Array Attributes	- Calculates the mean of array elements
	- Calculates the sum of array elements
	- Finds the minimum value in the array
	- Finds the maximum value in the array
	- Computes dot product of two arrays
		Example:
		`np.mean(array)`
		`np.sum(array)`
		`np.min(array`
		`np.max(array)`
		`np.dot(array_1, array_2)`

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